



Operating System Lab

CSE-3211

Assignment - 3

CPU Scheduling Algorithms to Find Turnaround Time and
Waiting Time

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1. Introduction

CPU scheduling is an important responsibility of an operating system. When multiple processes are ready to execute, the scheduler decides which process will receive the CPU next. In this experiment, five CPU scheduling algorithms were implemented in C and tested with five processes.

The implemented algorithms are:

- First-Come, First-Served (FCFS)
- Shortest Job First (SJF)
- Shortest Remaining Time Next (SRTN)
- Round Robin (RR)
- Lottery Scheduling

The main performance metrics used in this report are waiting time and turnaround time.

$$TurnaroundTime(TAT) = CompletionTime - ArrivalTime \quad (1)$$

$$WaitingTime(WT) = TurnaroundTime - BurstTime \quad (2)$$

For non-preemptive algorithms where all processes arrive at time 0, turnaround time is equal to completion time.

2. FCFS Scheduling

2.1 Input Set

The following five processes were used for First-Come, First-Served scheduling. All processes were considered to arrive at time 0.

Process	Burst Time (ms)	Arrival Time (ms)
P1	10	0
P2	4	0
P3	6	0
P4	2	0
P5	8	0

2.2 Result

In FCFS scheduling, processes execute in the same order in which they are entered. Therefore, the execution order is:

$$P1 - > P2 - > P3 - > P4 - > P5 \quad (3)$$

Process	Burst Time (ms)	Waiting Time (ms)	Turnaround Time (ms)
P1	10	0	10
P2	4	10	14
P3	6	14	20
P4	2	20	22
P5	8	22	30

Average Waiting Time = 13.20 ms
Average Turnaround Time = 19.20 ms

```

First-Come, First-Served (FCFS) Scheduling
Enter number of processes: 5
Enter burst time for P1: 10
Enter burst time for P2: 4
Enter burst time for P3: 6
Enter burst time for P4: 2
Enter burst time for P5: 8

Process Burst Time    Waiting Time    Turnaround Time
P1      10            0              10
P2       4           10              14
P3       6           14              20
P4       2           20              22
P5       8           22              30

Average waiting time: 13.20 ms
Average turnaround time: 19.20 ms

```

Figure 1: FCFS Scheduling Output

3. SJF Scheduling

3.1 Input Set

The same five processes were used for SJF so that the result could be compared with FCFS.

Process	Burst Time (ms)	Arrival Time (ms)
P1	10	0
P2	4	0
P3	6	0
P4	2	0
P5	8	0

3.2 Result

In SJF scheduling, the process with the smallest burst time is selected first. If burst times are equal, the earlier process is selected first.

$$P4- > P2- > P3- > P5- > P1 \quad (4)$$

Process	Burst Time (ms)	Waiting Time (ms)	Turnaround Time (ms)
P4	2	0	2
P2	4	2	6
P3	6	6	12
P5	8	12	20
P1	10	20	30

Average Waiting Time = 8.00 ms

Average Turnaround Time = 14.00 ms

3.3 Comparing with FCFS

Using the same process set, SJF produced a lower average waiting time than FCFS.

FCFS average waiting time = 13.20 ms

SJF average waiting time = 8.00 ms

Improvement = 13.20 - 8.00 = 5.20 ms

This shows that SJF can reduce average waiting time when shorter processes are allowed to execute before longer ones. However, long processes may suffer if shorter processes keep arriving repeatedly.

```

Shortest Job First (SJF) Scheduling
Enter number of processes: 5
Enter burst time for P1: 10
Enter burst time for P2: 4
Enter burst time for P3: 6
Enter burst time for P4: 2
Enter burst time for P5: 8

Execution order: P4 -> P2 -> P3 -> P5 -> P1

Process Burst Time      Waiting Time      Turnaround Time
P4       2              0                  2
P2       4              2                  6
P3       6              6                  12
P5       8             12                  20
P1      10             20                  30

Average waiting time: 8.00 ms
Average turnaround time: 14.00 ms

```

Figure 2: SJF Scheduling Output

4. SRTN Scheduling

4.1 Input Set

For SRTN, different arrival times were used to demonstrate preemption.

Process	Arrival Time (ms)	Burst Time (ms)
P1	0	8
P2	1	4
P3	2	9
P4	3	5
P5	4	2

4.2 Result

SRTN is the preemptive version of SJF. At each time unit, the ready process with the smallest remaining burst time is selected.

Process	Arrival	Burst	Completion	Waiting	Turnaround
P1	0	8	19	11	19
P2	1	4	5	0	4
P3	2	9	28	17	26
P4	3	5	12	4	9
P5	4	2	7	1	3

Average Waiting Time = 6.60 ms
Average Turnaround Time = 12.20 ms

```

Shortest Remaining Time Next (SRTN) Scheduling
Enter number of processes: 5
Enter arrival time and burst time for P1: 0 8
Enter arrival time and burst time for P2: 1 4
Enter arrival time and burst time for P3: 2 9
Enter arrival time and burst time for P4: 3 5
Enter arrival time and burst time for P5: 4 2

Execution log:
Time 0-1: P1
Time 1-2: P2
Time 2-3: P2
Time 3-4: P2
Time 4-5: P2
Time 5-6: P5
Time 6-7: P5
Time 7-8: P4
Time 8-9: P4
Time 9-10: P4
Time 10-11: P4
Time 11-12: P4
Time 12-13: P1
Time 13-14: P1
Time 14-15: P1
Time 15-16: P1
Time 16-17: P1
Time 17-18: P1
Time 18-19: P1
Time 19-20: P3
Time 20-21: P3
Time 21-22: P3
Time 22-23: P3
Time 23-24: P3
Time 24-25: P3
Time 25-26: P3
Time 26-27: P3
Time 27-28: P3

Process Arrival Burst Completion Waiting Turnaround
P1      0      8      19      11      19
P2      1      4       5       0       4
P3      2      9      28      17      26
P4      3      5      12       4       9
P5      4      2       7       1       3

Average waiting time: 6.60 ms
Average turnaround time: 12.20 ms

```

Figure 3: SRTN Scheduling Output

5. Round Robin Scheduling

5.1 Input Set

The same five burst times were tested with two different time quantum values.

Process	Burst Time (ms)
P1	10
P2	4
P3	6
P4	2
P5	8

5.2 Result

5.2.1 Quantum = 2ms

Process	Burst Time	Completion	Waiting Time	Turnaround Time
P1	10	30	20	30
P2	4	14	10	14
P3	6	22	16	22
P4	2	8	6	8
P5	8	28	20	28

Average Waiting Time = 14.40 ms

Average Turnaround Time = 20.40 ms

```

Round Robin Scheduling
Enter number of processes: 5
Enter burst time for P1: 10
Enter burst time for P2: 4
Enter burst time for P3: 6
Enter burst time for P4: 2
Enter burst time for P5: 8
Enter time quantum: 2

Execution order:
P1 (0-2)
P2 (2-4)
P3 (4-6)
P4 (6-8)
P5 (8-10)
P1 (10-12)
P2 (12-14)
P3 (14-16)
P5 (16-18)
P1 (18-20)
P3 (20-22)
P5 (22-24)
P1 (24-26)
P5 (26-28)
P1 (28-30)

Process Burst Time    Completion    Waiting Time    Turnaround Time
P1      10            30            20             30
P2       4            14            10             14
P3       6            22            16             22
P4       2             8             6              8
P5       8            28            20             28

Average waiting time: 14.40 ms
Average turnaround time: 20.40 ms

```

Figure 4: Round Robin Scheduling with time quantum 2 ms

5.2.2 Quantum = 4ms

Process	Burst Time	Completion	Waiting Time	Turnaround Time
P1	10	30	20	30
P2	4	8	4	8
P3	6	24	18	24
P4	2	14	12	14
P5	8	28	20	28

Average Waiting Time = 14.40 ms

Average Turnaround Time = 20.80 ms

```

Round Robin Scheduling
Enter number of processes: 5
Enter burst time for P1: 10
Enter burst time for P2: 4
Enter burst time for P3: 6
Enter burst time for P4: 2
Enter burst time for P5: 8
Enter time quantum: 4

Execution order:
P1 (0-4)
P2 (4-8)
P3 (8-12)
P4 (12-14)
P5 (14-18)
P1 (18-22)
P3 (22-24)
P5 (24-28)
P1 (28-30)

Process Burst Time      Completion      Waiting Time      Turnaround Time
P1      10             30              20                30
P2       4              8               4                 8
P3       6             24             18                24
P4       2             14             12                14
P5       8             28             20                28

Average waiting time: 14.80 ms
Average turnaround time: 20.80 ms

```

Figure 5: Round Robin Scheduling with time quantum 4 ms

5.3 Comparison

Round Robin gives each process a fair chance to execute by assigning a fixed time quantum. In this process set, the 2 ms quantum produced a slightly lower average waiting time than the 4 ms quantum.

Quantum 2 average waiting time = 14.40 ms

Quantum 4 average waiting time = 14.80 ms

Difference = 0.40 ms

A smaller quantum increases the number of context switches, while a larger quantum behaves more like FCFS. The best quantum depends on the process set and system requirements.

6. Lottery Scheduling

6.1 Input Set

The following five processes were used for all three Lottery Scheduling runs. The time quantum was 2 ms.

Process	Burst Time (ms)	Tickets
P1	10	10
P2	6	20
P3	4	5
P4	8	15
P5	5	25

6.2 Run 1

Process	Burst	Tickets	Completion	Waiting	Turnaround
P1	10	10	27	17	27
P2	6	20	29	23	29
P3	4	5	31	27	31
P4	8	15	33	25	33
P5	5	25	25	20	25

Average Waiting Time = 22.40 ms
 Average Turnaround Time = 29.00 ms

```

Lottery Scheduling
Enter number of processes: 5
Enter burst time and tickets for P1: 10 10
Enter burst time and tickets for P2: 6 20
Enter burst time and tickets for P3: 4 5
Enter burst time and tickets for P4: 8 15
Enter burst time and tickets for P5: 5 25
Enter time quantum: 2
Enter number of runs: 3

Run 1 execution log:
Time 0-2: draw=10/75, winner=P1
Time 2-4: draw=42/75, winner=P4
Time 4-6: draw=69/75, winner=P5
Time 6-8: draw=6/75, winner=P1
Time 8-10: draw=10/75, winner=P1
Time 10-12: draw=45/75, winner=P4
Time 12-14: draw=72/75, winner=P5
Time 14-16: draw=27/75, winner=P2
Time 16-18: draw=33/75, winner=P3
Time 18-20: draw=19/75, winner=P2
Time 20-22: draw=8/75, winner=P1
Time 22-24: draw=47/75, winner=P4
Time 24-25: draw=73/75, winner=P5
Time 25-27: draw=5/50, winner=P1
Time 27-29: draw=18/40, winner=P2
Time 29-31: draw=2/20, winner=P3
Time 31-33: draw=10/15, winner=P4

Process Burst   Tickets Completion   Waiting Turnaround
P1      10      10       27        17       27
P2       6      20       29        23       29
P3       4       5       31        27       31
P4       8      15       33        25       33
P5       5      25       25        20       25
Average waiting time: 22.40 ms
Average turnaround time: 29.00 ms

```

Figure 6: Lottery Scheduling Run 1

6.3 Run 2

Process	Burst	Tickets	Completion	Waiting	Turnaround
P1	10	10	33	23	33
P2	6	20	13	7	13
P3	4	5	23	19	23
P4	8	15	21	13	21
P5	5	25	7	2	7

Average Waiting Time = 12.80 ms
 Average Turnaround Time = 19.40 ms

```

Run 2 execution log:
Time 0-2: draw=57/75, winner=P5
Time 2-4: draw=68/75, winner=P5
Time 4-6: draw=35/75, winner=P3
Time 6-7: draw=57/75, winner=P5
Time 7-9: draw=23/50, winner=P2
Time 9-11: draw=14/50, winner=P2
Time 11-13: draw=12/50, winner=P2
Time 13-15: draw=21/30, winner=P4
Time 15-17: draw=19/30, winner=P4
Time 17-19: draw=18/30, winner=P4
Time 19-21: draw=22/30, winner=P4
Time 21-23: draw=15/15, winner=P3
Time 23-25: draw=6/10, winner=P1
Time 25-27: draw=10/10, winner=P1
Time 27-29: draw=5/10, winner=P1
Time 29-31: draw=4/10, winner=P1
Time 31-33: draw=3/10, winner=P1

Process Burst   Tickets Completion   Waiting Turnaround
P1      10       10        33          23        33
P2       6       20        13           7         13
P3       4        5        23          19         23
P4       8       15        21          13         21
P5       5       25         7           2          7
Average waiting time: 12.80 ms
Average turnaround time: 19.40 ms

```

Figure 7: Lottery Scheduling Run 2

6.4 Run 3

Process	Burst	Tickets	Completion	Waiting	Turnaround
P1	10	10	31	21	31
P2	6	20	11	5	11
P3	4	5	33	29	33
P4	8	15	29	21	29
P5	5	25	5	0	5

Average Waiting Time = 15.20 ms
 Average Turnaround Time = 21.80 ms

```

Run 3 execution log:
Time 0-2: draw=52/75, winner=P5
Time 2-4: draw=68/75, winner=P5
Time 4-5: draw=67/75, winner=P5
Time 5-7: draw=23/50, winner=P2
Time 7-9: draw=21/50, winner=P2
Time 9-11: draw=26/50, winner=P2
Time 11-13: draw=28/30, winner=P4
Time 13-15: draw=19/30, winner=P4
Time 15-17: draw=9/30, winner=P1
Time 17-19: draw=3/30, winner=P1
Time 19-21: draw=13/30, winner=P3
Time 21-23: draw=16/30, winner=P4
Time 23-25: draw=4/30, winner=P1
Time 25-27: draw=7/30, winner=P1
Time 27-29: draw=20/30, winner=P4
Time 29-31: draw=3/15, winner=P1
Time 31-33: draw=4/5, winner=P3

Process Burst  Tickets Completion  Waiting Turnaround
P1      10      10      31      21      31
P2       6      20      11       5      11
P3       4       5      33      29      33
P4       8      15      29      21      29
P5       5      25       5       0       5
Average waiting time: 15.20 ms
Average turnaround time: 21.80 ms

```

Figure 8: Lottery Scheduling Run 3

6.5 Discussion

Lottery Scheduling selects a process randomly using tickets. A process with more tickets has a higher chance of receiving the CPU, but the result is not fixed. The three runs used the same input and the same time quantum, yet they produced different completion times and average waiting times. This confirms the probabilistic behavior of Lottery Scheduling.

7. Conclusion

The experiment shows that each CPU scheduling algorithm has different behavior. FCFS is simple but may create long waiting times if a long process executes first. SJF improves average waiting time by giving priority to shorter burst times. SRTN extends SJF with preemption and improves responsiveness when shorter jobs arrive later. Round Robin is fair for time-sharing systems, but its performance depends on the selected quantum. Lottery Scheduling demonstrates proportional CPU sharing using random ticket selection, so repeated runs can produce different results.

All five algorithms were implemented in C, and the outputs were used to calculate completion time, waiting time, turnaround time, and average metrics.